

Among the most efficient photoelectrochemical reactions are those involving light absorption by wide-band-gap nanoparticles and molecular adsorbates, which result in the occurrence of electron-transfer reactions at the semiconductor surface that are the basis for photoelectric conversion in photovoltaic devices.^{2,8,9}

In contrast to studies on a bulk system, single-entity electrochemistry has emerged as a unique tool for exploring the and kinetic features of individual entities during stochastic collision measurements at a UME surface, avoiding any averaging and broadening effects.^{18,20}

For example, the electrocatalytic behavior of individual entities has been studied as they collide on a relatively inert UME, in which the stochastic collision electrochemical measurements offer direct electron-transfer information at interfaces and valuable evidence in studying the intrinsic electrochemical activities of an individual entity.^{18,23}

However, the key bottleneck in applying single-entity electrochemistry to photoelectrochemical measurement is the real-time recording of the ultrafast photoinduced electron-transfer data.³⁶

That is, the electron-transfer process of a photoelectrochemical reaction occurs on a sub-picosecond timescale and is electrochemically resolved because of the limitations of current electrochemical instruments.

Current photoinduced electron-transfer research is mainly based on ultrafast time-resolved transient spectroscopy for tracking the intermediate states of the electron-transfer process in photoelectrochemical reactions on picosecond femtosecond timescales.^{37,39}

Development and applications of the protocol A useful strategy for achieving in situ electrochemical measurement of electron transfer at the single-entity level is to fabricate a micron-thickness nanoparticulate TiO₂ /coated Au UME (UME) as an electron-transport electrode.

When individual N719@TiO₂ or N719@ZnO entities stochastically collide with TiO₂@Au UME, the electrons are into the nanoparticulate TiO₂ /lm.

Therefore, in our photoelectrochemical measurements, the rate-determining steps are the collision of a nanoparticle with TiO₂@Au UME and/or the electron diffusion process inside the TiO₂ /lm.

To elucidate the electron-transfer dynamics of the collision process, we studied the photoelectrochemical behaviors of dye-tagged oxide semiconductor nanoparticles colliding with a bare Au UME. No current signal was observed under in this case (Supplementary Fig. 1).

In this study, we performed single-entity photoelectrochemical experiments using a low-noise electrochemical at a sampling rate of 100 kHz.

As a result, the typical duration of electron transit in TiO₂ /lm was considerably increased, allowing the achievement successful real-time detection of single-entity photoelectrochemical behaviors.

Owing to a trap-mediated electron diffusion process in TiO₂ /lms, we observed submillisecond photocurrent associated with single-entity photoelectrochemical reactions.

Based on high-resolution photoelectrochemical measurements, our platform can be used for quantifying the electron-transfer process of a single event, enabling electron-transfer dynamics analysis at a semiconducting TiO₂ /lm.

It will certainly help to achieve an improved understanding of the electron-transfer process at the semiconductor where the photoelectrochemical reaction processes take place.

In addition, considering the extreme sensitivity of individual event signals, testing the effects of the filter and amplifier of low-noise electrochemical instrument should be performed as a best practice? standard in single-entity studies to minimize the electronics? effects on measurement results.^{48,49}

Higher light intensity causes an increase in current background noise due to local thermal drift. In this protocol, we step-by-step procedures for the fabrication of TiO₂@Au UMEs, the preparation of monodispersed dye-tagged oxide nanoparticle suspensions and the subsequent procedures to monitor the photoelectrochemical behaviors of individual in real time, process data and control single-nanoparticle events.

As demonstrated, this platform offers picoampere and sub-millisecond sensitivity in terms of the current and duration of individual collision events at a TiO₂@Au UME interface, allowing the quantitative measurement of photoinduced electrontransfer processes of a single entity.

CRITICAL All aqueous solutions are prepared with Milli-Q purified water (18.2 M Ω ·cm at 25 °C). CRITICAL All organic are dried by adding activated alumina.

CRITICAL Make fresh solution before each experiment.

CRITICAL STEP Adjust same temperature and burning time to avoid undesired bending of the Find individual entities in the sample at a low magnification ($\times 15$ k for the JEOL JEM-2100F TEM), spread the electron beam its widest, increase the magnification ($\times 600$ k for the JEOL JEM2100F TEM), and focus the electron beam slowly.

CRITICAL STEP The parameters used are highly dependent on the SEM instrument. ? TROUBLESHOOTING

CRITICAL STEP Avoid exposure of the sample to electron beam irradiation for >2 min in Plot experimental collision frequencies as a function of the entity concentration. (iii) Investigate the correlation the theoretical collision frequency (f_{NP}) and the nanoparticle concentration (C_{NP}) given by $f_{NP} = 4D_{NP}C_{NP}r_{UME}N_A$, where D_{NP} is the diffusion coefficient of nanoparticles, N_A is Avogadro's constant and r_{UME} is the radius of the TiO₂@Au UME. value of D_{NP} can be estimated using the Stokes-Einstein relationship by $D_{NP} = k_B T / (6 \pi \eta r_{NP})$, where k_B is Boltzmann's constant, T is temperature, η is the viscosity of the solution and r_{NP} is the radius of the N719@TiO₂ or N719@ZnO nanoparticle.

In this protocol, we have described detailed procedures for fabrication of an ultrasensitive photoelectrochemical which could be widely used for real-time monitoring of the electrontransfer process of a single entity in various systems.

Current-time curves of individual N719@TiO₂ entities colliding with a TiO₂@Au UME are collected in acetonitrile containing 10 mM LiI and 1 mM I₂ (Step 19A).

This indicates the photoinduced electron transfer does not occur without photoexcitation. Note that no current signal observed for the neat electrolyte solution without N719@TiO₂ entities, with illumination or without.

In addition, no current transient was observed for individual bare TiO₂ nanoparticles without the N719 dye in an amperometric current-time curve (Supplementary Fig. 8), demonstrating that the collected electrons result the photo-excited N719 moieties under visible-light illumination. Figure 5b shows the representative current traces that rise quickly and decay at a slower rate for individual N719@ TiO₂ entities.

Considering the electron recombination in a multiple-trapping TiO₂ /lm, the effective number of collected electrons a single N719 molecule per typical collision (N_e) at a TiO₂@Au UME from the measured transients was calculated by $N_e = Q / e$, where Q is the integrated charge of individual events, n is the number of electrons involved in the photoelectrochemical (1 for N719), e is the elementary charge and N₇₁₉ is the average number of N719 molecules modified onto the of a single TiO₂ nanoparticle surface (~ 600 ; Supplementary Fig. 9).

This allows us to calculate an N_e value of 51 ± 4 per N719 molecule (Fig. 5f), revealing the intrinsic electron-transfer of a single N719 molecule for a typical collision. In this measurement, the TiO₂ /lm is essential to the successful trapping the photoinduced electron-transfer process because no collision event could be observed for N719@TiO₂ entities at a bare UME (Supplementary Fig. 1).

This could be attributed to instantaneous collisions of individual entities beyond the measurement capability of electrochemical instrumentation or high energy dissipation for the transfer of electrons through the TiO₂ metal interface.⁵⁶

As a parallel study, we also investigated single-entity photoelectrochemistry for N719@ZnO nanoparticles in a DMSO/water solution at TiO₂@Au UMEs with different TiO₂ /lm thicknesses (Step 19B).

Owing to the varying diffusion and recombination processes of the photoinjected electrons in the different TiO₂ networks, thickness-dependent current traces were observed for individual N719@ZnO nanoparticle collisions to single-entity photocatalytic water oxidation (Fig. 6a).

In addition, the measured results of TiO₂ /lm thickness-dependent current traces further demonstrated that a trap-electron diffusion process was the rate-limiting step in our single-entity photoelectrochemical measurement.